

Artificial intelligence for joint angle monitoring and movement classification from Triboelectric Nanogenerator data

Semester Project

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June 14, 2026

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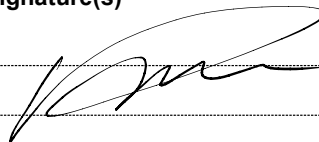
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Abstract

Triboelectric nanogenerators (TENGs) have emerged as a promising technology for self-powered wearable sensing due to their ability to convert mechanical motion into electrical signals without requiring an external power source. This work investigates the use of TENG signals for continuous knee angle estimation, activity recognition, and gait recognition, while exploring methods to improve generalization across individuals.

Using a dataset of synchronized TENG and optical motion capture recordings from eleven participants, two normalization approaches and several handcrafted feature sets were evaluated. In addition to the baseline amplitude-based features used in previous work, peak morphology and frequency-domain features were introduced to better capture the temporal structure of the TENG signal. Models were evaluated using both intra-participant and leave-one-participant-out inter-participant protocols, with TabPFN selected as the primary machine learning model.

For cycling knee angle estimation, excellent intra-participant performance was achieved, reaching ($R^2 = 0.985 \pm 0.015$) with a mean absolute error of ($2.35^\circ \pm 1.35^\circ$). Inter-participant regression performance improved substantially compared with previous work, achieving ($R^2 = 0.878 \pm 0.118$). Activity recognition achieved near-perfect performance in the intra-participant setting, while inter-participant results remained limited by subject variability. The proposed normalization approaches reduced amplitude differences between participants but provided little benefit for cycling regression and introduced label-dependent effects for activity classification. Finally, gait recognition achieved an impressive macro-F1 score of 0.985, demonstrating that TENG walking signals preserve highly distinctive subject-specific characteristics.

Overall, the results demonstrate that TENG-based sensors are highly effective for personalized motion monitoring and that feature engineering can substantially improve participant-independent knee angle prediction. However, inter-subject variability and data quality remain important challenges for robust deployment across diverse user populations.

Acknowledgements

This work was conducted under the supervision of Dr. Corin Otesteanu and Dr. Satyanranjan Bairagi, at the Biomedical and Mobile Health Technology (BMTH) Lab, ETH Zurich, under the direction of Prof. Dr. Carlo Menon. I am deeply grateful for their guidance, support, and valuable insights throughout this semester project. Furthermore, I am grateful to all members of the BMTH lab for their assistance and technical support.

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1 Introduction

Wearable sensor technology has evolved rapidly in the recent years, driven by the improvements in materials, electronics, and IoT technologies. These improvements were supported by the advancement in Artificial Intelligence methods that expanded the ability to extract meaningful information from sensor data, enabling automated and accurate analysis of human movement. With the increasing global demand for low cost and robust sensing systems, these developments led to the adoption of wearable technologies in a wide range of fields including healthcare, rehabilitation, sports performance analysis, occupational safety, and human-computer interaction.

Traditional camera based motion capture for activity monitoring, while highly accurate, comes with high costs, occlusion sensitivity, and privacy concerns for daily use [8]. Traditional wearable systems using IMUs provide a portable and low-cost alternative, however they are limited by high power consumption and rigid form [7] which reduces their long-term usability.

Triboelectric nanogenerators (TENGs) have gained increasing attention as a promising approach for self-powered wearable sensing systems by transforming mechanical motion into electrical energy through contact electrification and electrostatic induction [9]. In contrast to traditional sensing technologies, TENG-based sensors can harvest energy directly from natural body movements, removing the dependence on external power sources or batteries and enabling autonomous operation. The electrical outputs generated by TENGs, are strongly related to the mechanical deformations and motion patterns applied to them, making these devices particularly suitable for monitoring kinematic behavior. Due to this compatibility and their ability to be integrated into flexible, smart textiles with a wide range of materials, TENGs are increasingly being investigated not only for energy harvesting applications but also as wearable motion sensing devices capable of extracting detailed information about human movement and joint kinematics [10] [15].

A comparative study by Demircan et al. [4] evaluated TENG-based and IMU-based systems for HAR tasks and reported comparable classification performance while highlighting the self-powered characteristics of TENG sensors. In a broader review, Baburaj et al. [1] identified HAR, gesture recognition, and health monitoring as among the main application areas in which AI-assisted TENG systems have been investigated.

Gait analysis has emerged as one of the more frequently studied applications of wearable TENG sensors, largely due to the biomechanical information embedded in plantar pressure and foot motion signals. Zhang et al. [17] developed triboelectric smart socks combined with deep learning methods to perform gait analysis and user identification while simultaneously harvesting energy from walking. Their system employed an LSTM-based architecture and demonstrated high recognition performance across multiple users and activities. Zhang et al. [16] proposed a multi-sensor platform integrating TENG-enabled insoles with waist-mounted sensors to extract gait-related features for rehabilitation assessment using artificial neural networks. More recently, study [13] presented a flexible TENG-enabled insole system for self-powered gait monitoring in football-related activities, demonstrating the applicability of wearable TENG systems in sports biomechanics settings.

For joint angle estimation, the relationship between TENG output characteristics and mechanical deformation has motivated the use of these sensors for wearable motion monitoring. Wang et al. [14] demon-

strated a kinesiotape-based stretchable TENG placed over the knee joint and observed a repeatable voltage response across knee flexion angles, suggesting the feasibility of continuous angle monitoring without external power sources. More recently, Bairagi et al. [2] developed an all-fiber TENG (AF-TENG) integrated with a machine learning regression framework for direct joint angle estimation from sensor signals, reporting strong agreement between predicted and optically measured angles across multiple joints, including the knee, elbow, and wrist. Similarly, Gunawardhana et al. [5] investigated fabrigami- and electrospun nanofiber-based TENG structures for joint biomechanics monitoring and demonstrated their capability to capture both joint flexion magnitude and temporal movement characteristics relevant to biomechanical analysis.

1.1 Prior Work

Previous work in the Biomedical and Mobile Health Technology Lab established the foundation for TENG-based knee motion monitoring through the development of a wearable triboelectric nanogenerator (TENG) knee sleeve and the initial machine learning framework. Fabio Schaub [11] developed the sensor prototype and collected synchronized TENG and optical motion capture data from five participants performing cycling tasks, demonstrating the feasibility of continuous knee angle estimation from TENG voltage signals.

Building on this work, Pascal Spring [12] expanded the dataset by collecting data from six additional participants and introducing multiple movement tasks, including walking, squatting, knee bending, side leg raises, and cycling. In addition to continuous knee angle prediction, the study investigated human activity recognition from TENG signals and introduced an automated signal synchronization pipeline, extending the applicability of TENG-based sensing toward more diverse movement scenarios.

1.2 Focus of this Work

This work builds directly on Spring and Schaub's works by exploring new preprocessing methods as well as feature engineering with the goal of improving their continuous knee angle regression and activity classification results. Two new normalization approaches are investigated to improve subject invariance and facilitate better generalization to unseen individuals. In addition to regression and classification, current work evaluates gait recognition to assess the subject discriminative quality of TENG signals.

2 Methodology

2.1 Overview of the Sensor Design and Data Collection

The data used in this study were collected during previous measurements conducted by Schaub and Spring [11] [12]. The dataset combines recordings from five participants collected during Schaub’s study and six additional participants collected by Spring using the same experimental protocol and acquisition setup. During data acquisition, participants wore the TENG knee sleeve together with reflective markers for optical motion capture-based ground-truth estimation. Measurements included five movement tasks representing different movement categories: **biking**, **walking**, **squatting**, **knee bending**, and **side leg lifts**. TENG signals were recorded using a lightweight Bluetooth-enabled acquisition module, while knee kinematics were simultaneously captured using a multi-camera optical motion capture system. The resulting dataset was used as the basis for all analyses performed in this work.

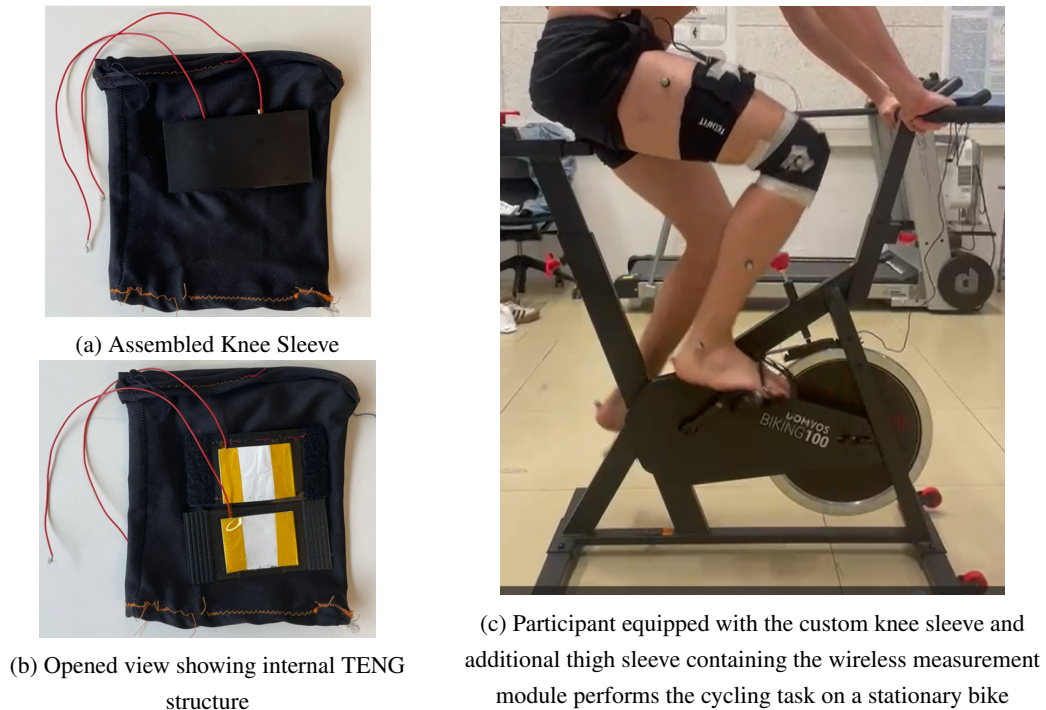


Figure 2.1: Wearable knee sleeve design and measurement setup [11].

2.2 Data Processing

The TENG data was preprocessed according to the approach proposed by Spring [12]. TENG signals were filtered using a second-order Butterworth low-pass filter with a 4 Hz cutoff frequency to suppress high-frequency noise and motion artifacts. Then, the TENG voltages are aligned with the OMC knee angle, using a peak detection-based approach. Corresponding peaks in both TENG and OMC signals are automatically detected. This is followed by a manual correction step where the user is able to crop the OMC signal, add or remove detected TENG peaks, as well as flip the polarity of the entire TENG signal with the goal of better matching the detected peaks of the two signals. Both signals are cut from the second detected peak to last after manual correction, before being aligned and downsampled accordingly. Following alignment and drift correction, TENG voltage signal and the ground truth knee angle from OMC were resampled to a common sampling rate of 50 Hz using linear interpolation. The entire processing was performed using Python(version 3.10.19). The OMC data was used already post-processed by Spring.

2.2.1 Data Quality Analysis

The TENG voltage signal was analysed using a bipolar SNR approach, since each movement cycle can produce both positive and negative voltage excursions. Peaks were detected separately in the voltage signal, $v(t)$, and in the inverted signal, $-v(t)$. The peak prominence threshold was defined as $3 \times 1.4826 \times \text{MAD}$ of the voltage signal. For the Biking task, a minimum peak distance of 0.40 s was used. Positive and negative peaks were then merged, and duplicate detections located very close to each other were removed by retaining the peak with the larger absolute deviation from the median voltage.

For each detected peak, the local signal region was defined using the absolute deviation from the median voltage, $|v(t) - \tilde{v}|$. Samples were classified as signal if they exceeded 30% of the local peak magnitude. Noise samples were defined as the remaining inactive samples, excluding a small window of ± 3 samples around each detected peak. Both signal and noise samples were median-centred before calculating SNR:

$$\text{SNR}_{\text{dB}} = 20 \log_{10} \left(\frac{\text{RMS}(v_{\text{signal}})}{\sigma_{\text{noise,robust}}} \right), \quad (2.1)$$

where $\sigma_{\text{noise,robust}}$ was estimated as $1.4826 \times \text{MAD}$ of the noise samples.

In addition to calculating SNR, all recordings were visually inspected before model training. SNR was used as a descriptive measure of signal quality, but it was not used as the sole exclusion criterion. This was because low-amplitude or irregular components of the signal may still contain movement-related information, while recordings with relatively high SNR may still contain artifacts such as double peaks, poor alignment, or non-task-related signal changes. Therefore, participant and task subsets were selected based on a combination of SNR values and visual inspection of the recorded voltage traces.

2.3 Normalization Approaches

One of the greatest challenges faced in human activity monitoring is the limitation of the model's generalization capabilities to new unseen individuals due to *inter-subject variability* [3]. The same activity is often performed differently across individuals due to the differences in physical characteristics such as body size, age, gender, anatomy, and movement execution style. In addition, wearable sensing systems introduce further variability through differences in sensor placement, attachment conditions, and signal amplitude scaling between users. To address this issue, two normalization approaches were proposed as a preprocessing step with the goal of reducing subject-specific variations while preserving motion-related information. By standardizing certain signal characteristics across participants, these approaches aim to improve model robustness and enhance generalization to previously unseen individuals.

2.3.1 Approach 1 - Voltage-Angle Fit Approach

The first approach exploits the physical relationship between the TENG voltage output and the underlying joint motion. Because the sensor is attached over the knee, its voltage V varies approximately linearly with the knee flexion angle $\theta(t)$ within the range of motion of an activity. For each participant p a linear model is fitted by least squares across all of their data for a given task,

$$V_p(t) = a_p \theta(t) + b_p. \quad (2.2)$$

The slope $a_p = dV/d\theta$ (in $V/^\circ$) captures the participant-specific *sensor sensitivity*, which is how strongly the voltage responds to a unit change in angle, and the intercept b_p captures the *DC offset* of the signal at $\theta = 0$. Variations in a_p and b_p across participants thus reflect exactly the subject or condition specific variations we wish to remove.

Each participant's signal is then mapped onto the scale of a chosen reference participant (ref) by

$$V_{\text{scaled}}(t) = \frac{a_{\text{ref}}}{a_p} (V_p(t) - b_p) + b_{\text{ref}}. \quad (2.3)$$

Subtracting b_p removes the individual DC offset, the ratio a_{ref}/a_p rescales the signal so that one degree of knee motion produces the same voltage change as for the reference participant, and adding b_{ref} places the result on the reference participant's absolute scale. Because this approach calibrates against the measured joint angle, it aligns the signals in a motion-aware manner, but it requires synchronized ground-truth angle data and is therefore only applicable to cases for which such reference angles are available.

2.3.2 Approach 2 - Voltage Amplitude Range Approach

The second approach removes subject-specific scaling using only statistics of the voltage signal itself, without requiring any angle reference. For each participant and task, the overall signal amplitude is estimated as

$$\Delta V_p = V_{p,\text{max}} - V_{p,\text{min}}, \quad (2.4)$$

and the DC offset is taken as the signal mean $b_p = \overline{V_p}$. To avoid sensitivity to isolated outliers, $V_{p,\text{min}}$ and $V_{p,\text{max}}$ are computed as the 1st and 99th percentiles of the voltage rather than its absolute extremes. The signal is then matched to the reference participant through

$$V_{\text{scaled}}(t) = \frac{\Delta V_{\text{ref}}}{\Delta V_p} (V_p(t) - b_p) + b_{\text{ref}}, \quad (2.5)$$

which centres each participant's signal, rescales its dynamic range to match that of the reference, and re-centres it on the reference offset. Unlike the first approach, this normalization does not depend on matching angle ranges or on any ground-truth motion data, making it broadly applicable across all activities, including those for which no joint-angle reference exists.

2.3.3 Calibration

A potential limitation of the full-recording normalization procedure is that the participant-specific parameters are estimated using all samples from a task. To test whether similar normalization could be achieved under a more deployment-realistic setting, a calibration-window variant was evaluated for Approach 1 and 2. In this analysis, the linear relationship between TENG voltage and knee angle (approach 1) and the voltage (approach 2), were estimated using only a short initial segment of each trial. The first 2 s of each trial were discarded to remove startup transients, and the first 10% of the remaining stable samples were used to estimate the scaling values and offsets. These calibration-derived parameters were then applied to the full voltage waveform using the same equations as in the main normalization procedures described above.

2.4 Evaluation Protocols

Three separate machine learning tasks were evaluated: regression for continuous knee angle prediction, classification for activity recognition, classification for subject gait recognition. For regression and classification tasks, intra-participant and inter-participant evaluation protocols were employed to assess personalized and generalized model performance, respectively. In all settings, the above described preprocessing procedure was applied identically, while normalization and the choice of approach were optional. The voltage signal was segmented into windows of varying lengths and strides and the desired feature set was extracted per window. All features were z-score standardized, with the normalization statistics computed exclusively on the training data and then applied to the held-out validation and test sets to avoid information leakage. These features are then fed into the selected model for training. The window size and stride combination was optimized along with other hyperparameters during the validation stage. This workflow matches the previous works [11] and [12] for comparability.

2.4.1 Intra-participant evaluation

Intra-participant evaluation assesses model performance under *personalized settings*, where training and testing are performed separately for each participant. In this setup, the model is evaluated on unseen samples from the same individual, allowing the analysis of how well participant-specific patterns can be learned while minimizing the influence of inter-subject variability.

For the intra-participant evaluation, models were trained and tested separately for each participant to analyze personalized performance. Each participant's processed dataset was first split chronologically into 70% training, 15% validation, and 15% testing subsets. The temporal order of the data was preserved to avoid any information leakage between sets. The windowing procedure described above was applied independently within each subset, ensuring that no overlapping windows spanned across partition boundaries.

Hyperparameters are selected via grid search on the validation set (minimizing MAE), after which the scaler and model are refit on the combined training and validation data before final evaluation on the held-out test split.

2.4.2 Inter-participant evaluation

Inter-participant evaluation assesses a model's ability to *generalize to unseen individuals* by separating participants between training and testing. Unlike intra-participant evaluation, this setting evaluates whether participant-independent patterns can be learned despite inter-subject variability, making it a more realistic representation of real-world deployment scenarios.

To assess generalization to unseen individuals, a leave-one-participant-out (LOPO) scheme was adopted. In each fold, one participant was held out as the test set, while the remaining participants were used for training. Hyperparameters were tuned using participant-level cross-validation within the training group: each training participant served as the validation set once, and the best hyperparameters were selected based on the average performance across these folds. The final model was then retrained on all training participants using the selected hyperparameters. This procedure was repeated for each participant, and the reported metrics represent the mean and standard deviation across all LOPO folds.

2.4.3 Regression

For the knee angle prediction task, multiple temporal configurations were investigated using window lengths of 100 ms, 200 ms, 500 ms, 1000 ms, and 2000 ms combined with stride values of 10 %, 30 %, 50 %, 70 %, and 90 %. Results in the following chapter are reported using overlap percentages, where overlap is defined as 100% - stride.

Each window was assigned a target value corresponding to the average ground-truth knee angle computed within a small region of ± 5 samples around the center of the window. Model performance was assessed using the coefficient of determination (R^2) and mean absolute error (MAE), with MAE reported in degrees. Definitions of these metrics are provided in Appendix A.1.

2.4.4 Classification

For activity recognition across the five activities (bike, walk, squat, knee bend, and side lift), window sizes of 1000 ms, 2000 ms, and 3000 ms were evaluated with the same stride values as described above. Classification performance was assessed using balanced accuracy and the macro-averaged F1-score to account for the class imbalance. The formulas for these metrics are provided in Appendix A.1.

2.4.5 Gait Recognition

The Gait recognition task was performed on walk activity across seven participants that had complete walk data. Similarly to the classification task, window sizes of 1000 ms, 2000 ms, and 3000 ms were evaluated with the same stride values as in regression task. The goal of this task was to predict the participant from their gait (walking) data. Results were evaluated with the balanced accuracy and macro-averaged F1 metrics.

2.5 Machine Learning

2.5.1 Feature Extraction

After preprocessing, the TENG voltage signals were divided into fixed-length temporal windows according to the selected window size and stride. For each window, handcrafted features were extracted from the voltage signal and used as input to the machine learning models. Three feature configurations were investigated in this work. The features making up these configurations are divided into three categories listed in Table 2.1.

voltage amplitude	peak morphology	frequency
v_{center}	n_{peaks}	f_{dom}
v'_{center}	Peak rate (peaks/s)	Spectral centroid
v_{min}	Mean peak height	Spectral entropy
v_{max}	Std. peak height	Low-band power ratio
v_{mean}	Mean prominence	Strongest autocorrelation peak
	Std. prominence	
	Mean peak width	
	Std. peak width	
	Mean inter-peak interval	
	Std. inter-peak interval	
	Mean peak sharpness	
	Std. peak sharpness	

Table 2.1: Overview of the evaluated feature sets and their extracted features.

The **Base** feature set consists only of the voltage amplitude features and follows the approach used by Spring [12]. For each window, five amplitude-descriptive features were extracted: the voltage value at the center of the window v_{center} , the temporal derivative at the center of the window v'_{center} , the minimum voltage v_{min} , the maximum voltage v_{max} , and the mean voltage v_{mean} . These features capture both the instantaneous voltage response and the overall amplitude range of the signal within the window.

Base+Peak feature set is made up of both the voltage amplitude and peak morphology features. The goal introducing peak morphology features was to better characterize the repetitive structure of the TENG signal and these features were extracted using a peak detection-based approach.

Before peak detection, each window was centered by subtracting its median. Peaks were then detected using constraints on minimum peak distance, prominence, and height. From the detected peaks, twelve morphology-related features were extracted: the number of detected peaks (n_{peaks}), peak rate (peaks/s), mean and standard deviation of peak heights, mean and standard deviation of peak prominences, mean and standard deviation of peak widths, mean and standard deviation of inter-peak intervals, and mean and standard deviation of peak sharpness. Peak sharpness was defined as the ratio between peak prominence and peak width. These features were selected because TENG signals during cyclic or repetitive movements often contain distinct voltage peaks corresponding to contact-separation events of the sensor layers. Therefore, peak morphology can provide information about movement rhythm, signal consistency, and deformation intensity.

The **Base+Peak+Freq** is the largest feature set, combining all feature types. Frequency-domain features were introduced with the goal of capturing periodicity and spectral structure. For each window, the DC component was removed and the frequency spectrum was computed using the fast Fourier transform. Five scale-invariant spectral features were extracted: the dominant frequency f_{dom} , spectral centroid, spectral entropy, low-band power ratio, and the strongest secondary autocorrelation peak. The low-band power ratio was computed as the proportion of spectral power between 0.5–2 Hz relative to the total power in the 0.5–5 Hz movement band. The autocorrelation feature corresponds to the magnitude of the strongest non-zero-lag autocorrelation peak and serves as a measure of signal periodicity. These features were designed to distinguish continuous periodic movements, such as biking and walking, from more intermittent repetition-based movements, such as squats, knee bends, and side lifts. Since these features are based primarily on normalized frequency distributions rather than absolute voltage amplitudes, it was proposed that they may also improve robustness to inter-participant differences in signal scale.

All extracted features were subsequently standardized during model evaluation using statistics computed only from the training data before being provided to the selected machine learning model.

2.5.2 TabPFN

This work primarily focused on the TabPFN (Tabular Prior-data Fitted Network) model, as it demonstrated the strongest overall performance in the previous work by Spring [12] across both regression and classification tasks. Building upon these findings, TabPFN was selected as the main model for further investigation and optimization in this work.

TabPFN is a transformer-based foundation model designed specifically for tabular machine learning tasks. Unlike conventional machine learning methods that learn task-specific representations entirely from the available training data, TabPFN is pretrained on a large collection of synthetically generated tabular datasets sampled from diverse data-generating processes. Through this pretraining procedure, the model learns a prior over tabular prediction problems and can perform efficient inference on previously unseen datasets [6].

The use of TabPFN is particularly suitable for this work due to the characteristics of the dataset and feature representation. The low-dimensional hand-crafted features extracted from fixed temporal windows naturally creates a structured tabular data. This structure is perfect for TabPFN, which was explicitly developed for such input forms. Furthermore, the small number of participants and the small dataset size appears to be a challenge for traditional deep learning approaches that rely on a large number of data to learn good descriptions. TabPFN on the other hand was shown to perform strongly on small and medium sized tabular datasets.

For classification tasks, the `TabPFNClassifier (v2.5)` implementation was employed for both activity recognition and gait recognition experiments. Window-level feature vectors were provided as inputs and the model produced class probability estimates for each target class. For continuous knee angle prediction, the `TabPFNRegressor (v2.5)` implementation was used, where the extracted feature vectors

served as inputs and the corresponding knee angles were used as regression targets. Both implementations were evaluated using the intra-participant and inter-participant protocols described above to enable direct comparison with previous studies and baseline models.

Due to the transformer-based architecture and computational requirements of TabPFN, training and inference were performed using CUDA-enabled GPU acceleration and Python (version 3.11.6).

3 Experiments and Results

3.1 Data Processing and Analysis

3.1.1 SNR Results for Bike Data

Bipolar SNR was calculated for the bike task for 10 participants, with two trials per participant. The per-participant median SNR ranged from 12.67 dB to 29.55 dB. Across participant medians, the overall median SNR was 17.62 dB and the mean was 18.35 dB.

Participant	Median SNR [dB]	IQR [dB]
P1	29.55	0.10
P3	22.93	0.49
P4	22.30	0.95
P5	21.84	0.81
P11	18.73	1.46
P9	16.52	1.25
P2	13.48	0.32
P10	12.77	0.28
P6	12.76	0.47
P8	12.67	0.38

Table 3.1: Signal-to-noise ratio (SNR) statistics for each participant only for Bike activity. Median SNR and interquartile range (IQR) are reported across two trials.

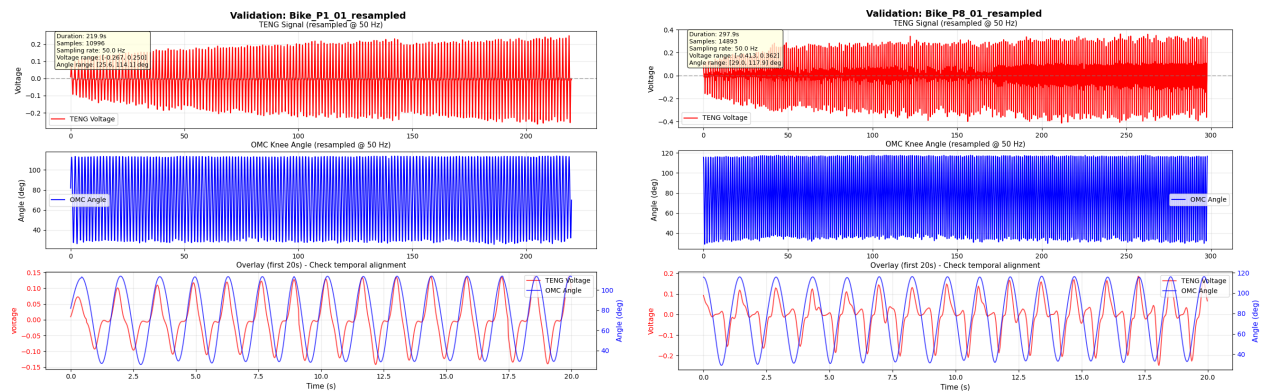


Figure 3.1: Comparison of aligned P1 TENG signal and P8 TENG signal for Biking. Left: P1. Right: P8.

Based on the SNR analysis and visual inspection of the voltage signals, selected participant groups were used for the task-specific model analyses. The selected groups were chosen to include recordings with consistent task-related signal patterns and without major artifacts that would make the data unreliable for model training.

3.1.2 Normalization

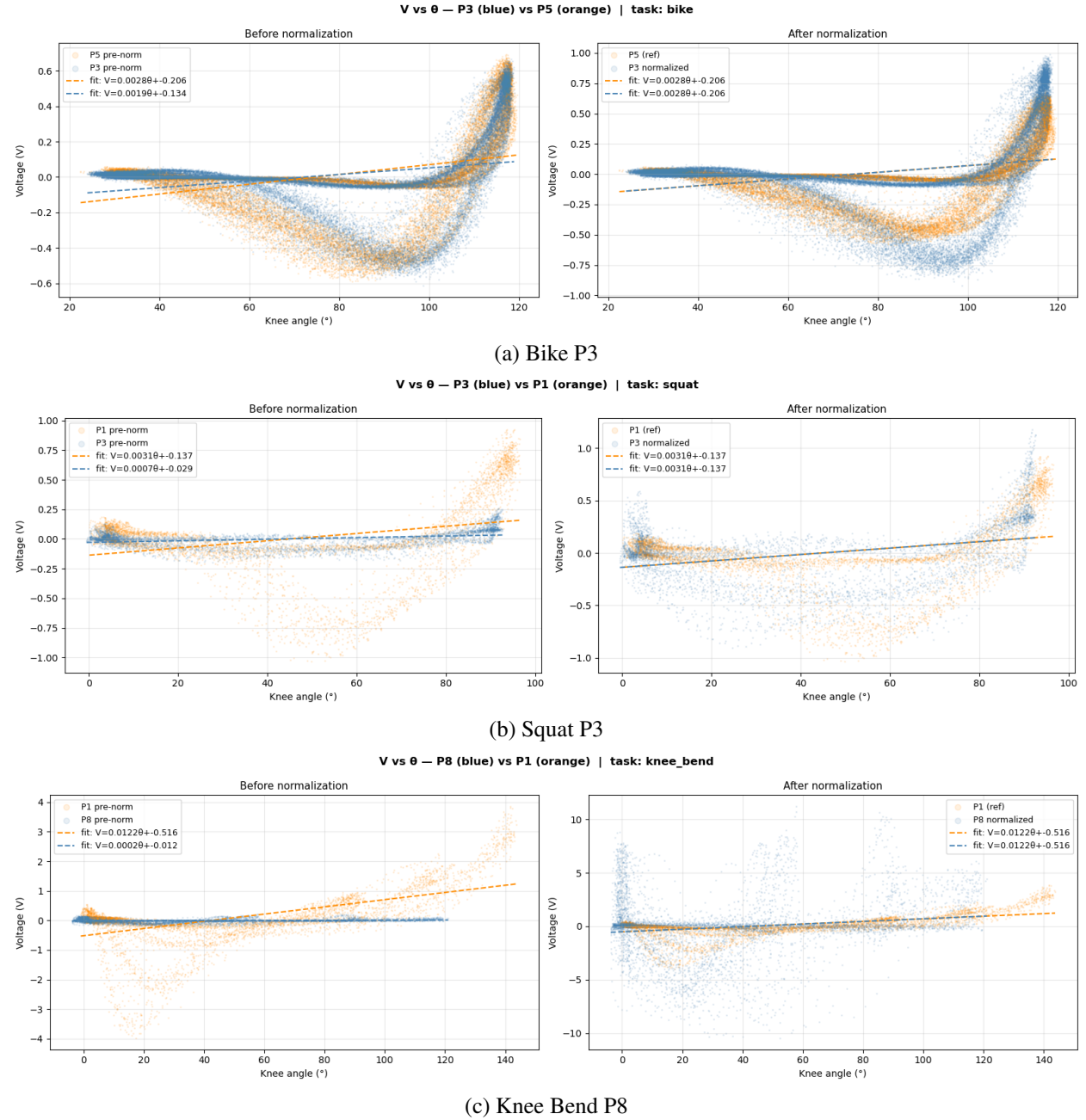


Figure 3.2: Normalization approach 1 before and after scatter plots.

Figure 3.2 demonstrates the linear fit of selected participant and the reference participant after normalization via Approach 1. For Bike activity, the inter-subject variance is limited and the linear fit between knee angle and TENG voltage differs slightly among participants. The change in linear fit after normalization for participant P3 - Bike activity can be seen in Figure 3.2 (a) where P5 is taken as the reference participant. Effects of normalization can be seen more drastically in Squat and Knee Bend activities. Figure 3.2 (b) and 3.2 (c) demonstrate this effect for P3 - Squat and P8 - Knee Bend where P1 is taken as reference participant of normalization. Although the effect of reference choice is not well demonstrated with these scatter plots, it is mentioned in the following sections.

Figure 3.3 shows a bar chart of per participant voltage ranges before and after normalization via Approach 2. The effect of normalization via Approach 2 is similar across different activities.

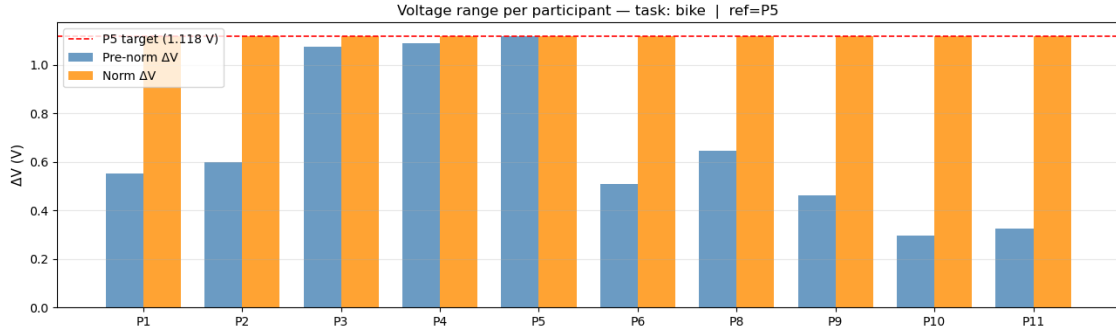


Figure 3.3: Normalization approach 2, voltage range bar chart before and after normalization for all participants.

3.1.3 Calibration

For Approach 1, the normalized waveform obtained from the calibration-window fit was compared sample-by-sample with the waveform obtained from the full-recording fit. Mean absolute error (MAE) was used as the primary comparison metric. For the Biking task, using P5 as the reference participant; P11, which had near-zero fitted slope was excluded because the linear voltage-angle relationship was not meaningful. The procedure was done for all activity types and, participants that do not have complete data for that activity were excluded. A similar procedure was followed for Approach 2. All participants with complete data were included in the analysis for each activity type and outliers as in the case described above were not observed.

Task	n	MAE Mean $\pm$ Std	Median
Bike	9	0.0278 ± 0.0327	0.0179
Knee bend	9	0.0438 ± 0.0815	0.0185
Squat	10	0.0414 ± 0.0227	0.0415
Walk	7	0.0154 ± 0.0205	0.0100

(a) Approach 1

Task	n	ΔV Ratio	MAE Mean $\pm$ Std
Bike	10	0.87 ± 0.22	0.036 ± 0.027
Walk	7	0.86 ± 0.16	0.012 ± 0.011
Squat	10	0.92 ± 0.14	0.014 ± 0.006
Knee bend	9	0.95 ± 0.07	0.018 ± 0.011
Side lift	10	1.11 ± 0.43	0.001 ± 0.001

(b) Approach 2

Table 3.2: Calibration-window robustness results comparing normalization parameters estimated from a 10% calibration window against those estimated from the full recording.

These results indicate that participant-specific linear scaling parameters or the voltage range and DC offset parameters do not necessarily require access to the full task recording. Instead, they can be approximated from a short initial calibration segment. This reduces the concern that the normalization depends on

statistics computed from the entire recording, including samples that would not be available at the beginning of a deployment session.

3.2 Regression

Regression experiments reported in this section were only done for **Bike** activity for better comparison with the previous works. After preprocessing and data inspection, participants **P1, P2, P3, P4, P5, P6, and P8** were chosen to be the most suitable. For all evaluations reported in this section, **Base+Peak+Freq** feature set was used.

3.2.1 Intra-Participant

In the intra-participant setting, TabPFN regressor was trained and evaluated individually for each participant, using solely that participant’s unnormalized data. Table 3.3 shows the best performance averaged across participants for the selected configuration. Using Base+Peak+Freq feature set on unnormalized data, TabPFN regressor achieved an impressive R^2 value of 0.985 ± 0.015 with 2.35 ± 1.35 MAE[°]. Table 3.4 is a breakdown of per participant best results achieved in this setting. Lastly, Figure 3.4 shows a 10 second example of the knee angle prediction made by the model for participant P6. This figure demonstrates the success of the model as the prediction aligns closely with the ground truth.

Model	Window [ms]	Overlap [%]	R^2	MAE [°]	RMSE [°]
TabPFN	1000	90	0.985 ± 0.015	2.35 ± 1.35	3.11 ± 1.65

Table 3.3: Intra-participant regression metrics for the best hyperparameter configuration achieved with TabPFN regressor evaluated on ‘biking’ activity data.

Participant	Window [ms] / Overlap [%]	R^2	MAE [°]	RMSE [°]
P1	1000 / 70	0.993	1.83	2.39
P2	1000 / 90	0.993	1.68	2.27
P3	500 / 90	0.974	3.39	4.73
P4	1000 / 30	0.996	1.44	1.82
P5	500 / 90	0.961	4.78	5.67
P6	1000 / 90	0.989	1.67	2.35
P8	1000 / 90	0.996	1.29	1.69

Table 3.4: Best intra-participant regression results per participant, obtained with TabPFN regressor.

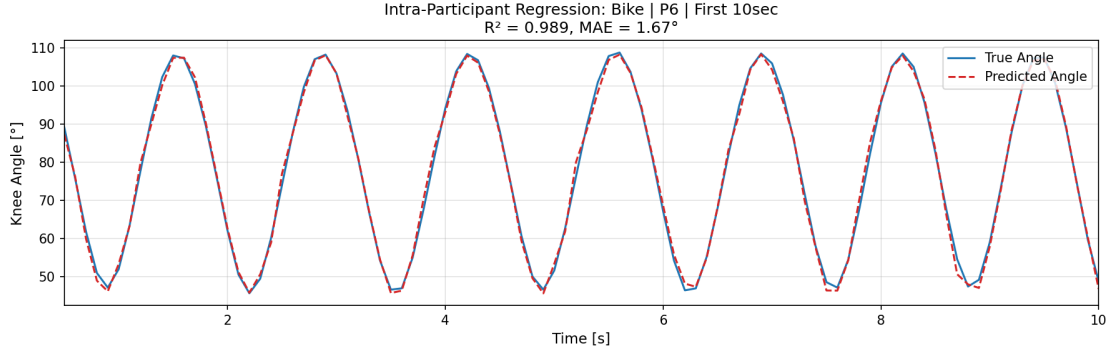


Figure 3.4: Example 10s prediction from P6 best combination, compared to ground truth angle

3.2.2 Inter-Participant

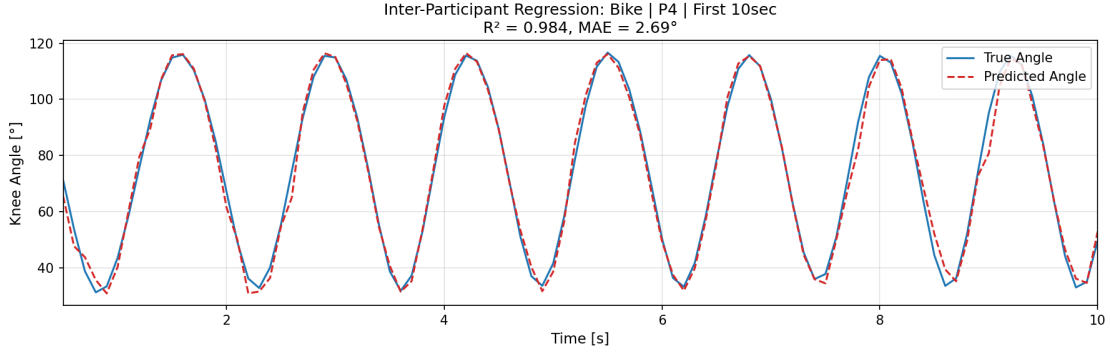
In the inter-participant setting, the TabPFN regressor was trained using a leave-one-participant-out (LOPO) strategy, where one participant was held out for testing and the remaining participants were used for training and inner participant-level cross-validation. Since the model was evaluated on participants whose data were not seen during training, this setting is more challenging than the intra-participant case. Optionally, the data were normalized using Approach 1 before feature extraction and the Base+Peak+Freq feature set was used. As shown in Table 3.5, the best average performance was achieved with a 1000 ms window and 30% overlap, resulting in an R^2 of 0.878 ± 0.118 , MAE of $6.84 \pm 2.97^\circ$, and RMSE of $8.29 \pm 3.34^\circ$. It can also be seen that near identical metrics were achieved with unnormalized data. Table 3.6 shows that performance varied across held-out participants: P3, P4, and P5 generalized well with $R^2 > 0.95$, while P6 was the most difficult participant, with an R^2 of 0.698 and MAE of 9.16° . Overall, these results indicate that inter-participant knee angle regression is feasible, but less stable than intra-participant regression due to participant-specific signal differences.

Model	Normalization	Window [ms]/Overlap [%]	R^2	MAE [$^\circ$]	RMSE [$^\circ$]
TabPFN	None	1000/90	0.878 ± 0.133	6.70 ± 2.96	8.17 ± 3.44
TabPFN	Approach 1	1000/30	0.878 ± 0.118	6.84 ± 2.97	8.29 ± 3.34

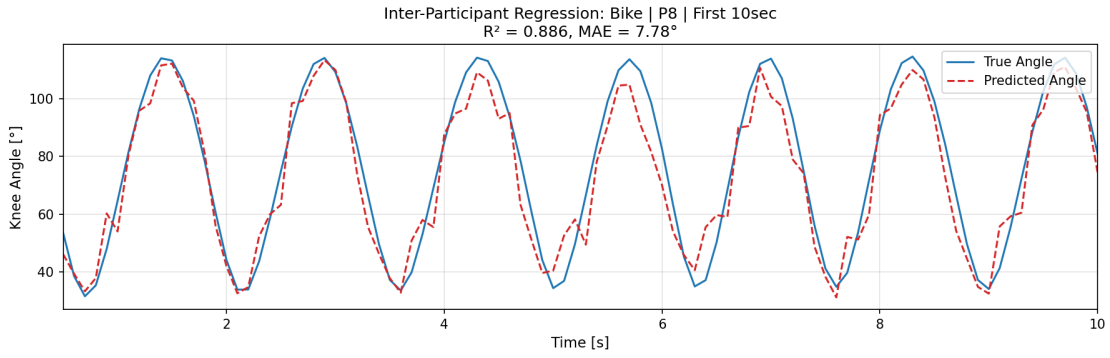
Table 3.5: Inter-participant regression performance using TabPFN regressor with and without normalization.

Participant	Window [ms] / Overlap [%]	R^2	MAE [$^\circ$]	RMSE [$^\circ$]	CV MAE [$^\circ$]
P1	1000 / 30	0.910	7.57	8.95	6.71
P2	1000 / 10	0.872	8.46	9.49	6.96
P3	1000 / 30	0.957	4.94	6.21	7.13
P4	1000 / 90	0.984	2.69	3.66	7.90
P5	2000 / 90	0.975	3.49	4.58	8.93
P6	2000 / 30	0.698	9.16	11.90	8.32
P8	1000 / 90	0.886	7.78	9.25	6.71

Table 3.6: Best inter-participant regression performance for each held-out participant using TabPFN and normalization Approach 1.



(a) P4, best inter-participant configuration



(b) P8, best inter-participant configuration

Figure 3.5: Example predictions compared to ground truth knee angle.

The example images above in Figure 3.5 show some of the knee angle predictions against the ground truth. The figure (b) is particularly representative since it is from a model (P8 held-out) with similar metrics to overall regression performance.

3.3 Classification

3.3.1 Intra-Participant

For the intra-participant classification task, TabPFN classifier was evaluated individually for each participant on unnormalized data. Evaluation was done in two groups due to missing data considerations. A classification task with all five activities (biking, walking, squatting, knee bending, and side leg lifts) was evaluated on P1, P2, P3, P4, P5, P6, P10 data. Another classification task on P1, P2, P3, P4, P5, P6, P8 with 4 activities excluding walking. For all cases, Base+Peak+Freq feature set was used.

As shown in Tables 3.7 and 3.8, the classifier achieved excellent performance in both scenarios. For the five-activity task, a balanced accuracy of 0.982 ± 0.038 , an F1 macro score of 0.983 ± 0.038 , and an F1 weighted score of 0.993 ± 0.015 were obtained. The corresponding precision and recall values were also high, reaching 0.985 ± 0.036 and 0.982 ± 0.038 , respectively. Similarly, for the four-activity task excluding walking, the model achieved a balanced accuracy of 0.978 ± 0.047 , an F1 macro score of 0.979 ± 0.047 , and an F1 weighted score of 0.991 ± 0.017 , with precision and recall values of 0.981 ± 0.047 and 0.978 ± 0.047 , respectively. The high scores across all metrics indicate that the extracted features

provide strong discrimination between activities for a given participant, while the low standard deviations suggest consistent performance across the evaluated subjects.

Model	Window [ms] / Overlap [%]	Balanced Acc.	F1 Macro	F1 Weighted
TabPFN	3000 / 30	0.982 ± 0.038	0.983 ± 0.038	0.993 ± 0.015

Table 3.7: Intra-participant classification performance using TabPFN and the combined Base+Peak+Freq feature set. All 5 activities, P1-6,10

Model	Window [ms] / Overlap [%]	Balanced Acc.	F1 Macro	F1 Weighted
TabPFN	3000 / 10	0.978 ± 0.047	0.979 ± 0.047	0.991 ± 0.017

Table 3.8: Intra-participant classification performance using TabPFN and the combined Base+Peak+Freq feature set. 4 activities excluding walking, P1-6,8

3.3.2 Inter-Participant

For inter-participant classification, TabPFN classifier was trained using a leave-one-participant-out (LOPO) strategy, where one participant was held out for testing and the remaining participants were used for training and inner participant-level cross-validation. All results were obtained from participant group P1-6,10 data and using Base+Peak+Freq feature set. Different normalization approaches and different normalization references were applied and the results are listed in Table 3.9.

The normalized conditions show substantially higher performance than the non-normalized baseline. However, these results should be interpreted as optimistic. The normalization transforms were computed and applied separately for each activity, meaning that the preprocessing applied to each window depended on its true activity label. Therefore, the normalized results do not correspond to strict label-agnostic LOPO classification. They are better interpreted as performance under a supervised, task-specific normalization setting. The true LOPO performance without label leakage is a balanced accuracy of 0.651 ± 0.229 and an F1 Macro score of 0.627 ± 0.259 , which is obtained without normalization.

Model	Normalization	Reference	Ws[ms] / Ov[%]	Balanced Acc.	F1 Macro	F1 Weighted
TabPFN	None	None	3000 / 70	0.651 ± 0.229	0.627 ± 0.259	0.661 ± 0.307
TabPFN	Approach 1	P5	3000 / 90	0.726 ± 0.141	0.679 ± 0.176	0.786 ± 0.166
TabPFN	Approach 2	P5	3000 / 30	0.811 ± 0.106	0.803 ± 0.115	0.867 ± 0.117
TabPFN	Approach 1	P1	3000 / 30	0.838 ± 0.077	0.795 ± 0.099	0.793 ± 0.125
TabPFN	Approach 2	P1	3000 / 30	0.890 ± 0.082	0.875 ± 0.095	0.857 ± 0.141

Table 3.9: Inter-participant classification performance using TabPFN for different normalization strategies and reference participants. Results are reported as mean $\pm$ standard deviation across the seven LOPO folds.

Although both approaches rely on task-specific normalization and therefore introduce a class label leakage, Approach 2 consistently outperformed Approach 1. When the confusion matrices in Figure 3.6 are inspected for Approach 1 and 2 normalization with same reference participant, the improvement is concentrated primarily in squat, knee bend, and walking activities, whereas biking exhibits only a modest change. This suggests that the two normalization strategies affect the discriminative information contained in the TENG signals differently. A possible explanation is discussed in Section 4

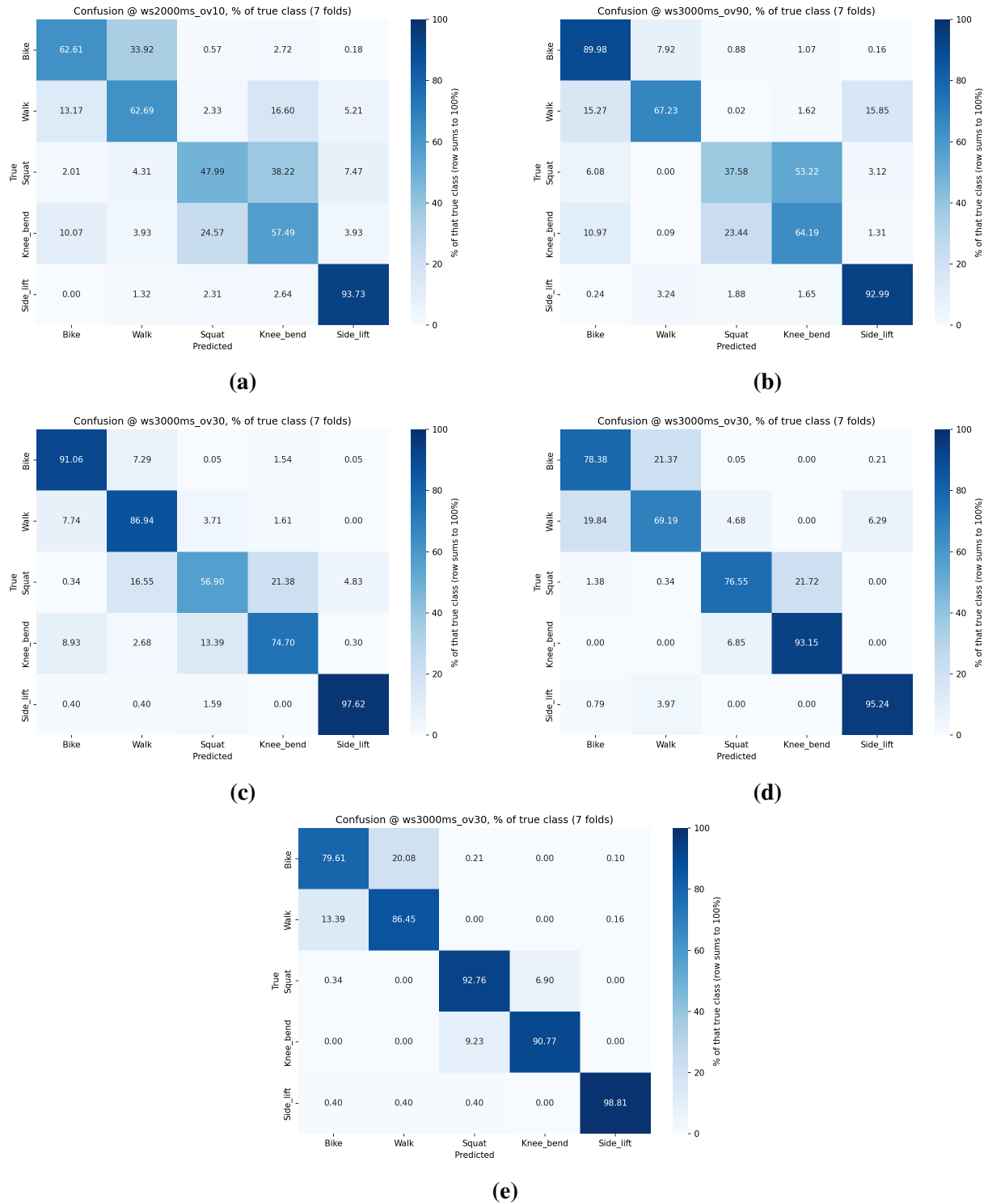


Figure 3.6: Comparison of confusion matrices for different normalization approaches. (a) No normalization, (b) Approach 1 with P5 reference, (c) Approach 2 with P5 reference, (d) Approach 1 with P1 reference, and (e) Approach 2 with P1 reference.

The confusion matrices also demonstrate the importance of the choice of reference participant. It can be observed that when the reference is changed from P5 to P1 (for either approach) there is great improvement

in classification of squat and knee bend activities. This occurs because the knee bend and squat data of participant P5 is more noisy and includes artifacts such as double TENG voltage peaks that was discussed in 4.6.2.

3.4 Feature Set Comparison

Table 3.10 shows the best intra-participant classification results for three different feature sets. These results were obtained by using unnormalized data of all five activities from participants P1-6,10. The Base feature set with only voltage amplitude features achieved the lowest F1 Macro score of 0.915 ± 0.086 while the extended Base+Peak feature set containing both amplitude and peak morphology features increased the F1 Macro score to 0.967 ± 0.045 . The largest and most descriptive Base+Peak+Freq feature set with additional frequency features returned the highest F1 Macro score of 0.981 ± 0.023 as expected.

Feature Set	Window [ms]/ Overlap [%]	Balanced Acc.	F1 Macro	F1 Weighted
Base	2000/90	0.916 ± 0.083	0.915 ± 0.086	0.944 ± 0.056
Base+Peak	2000/10	0.964 ± 0.048	0.967 ± 0.045	0.981 ± 0.023
Base+Peak+Frequency	3000/30	0.982 ± 0.038	0.983 ± 0.038	0.993 ± 0.015

Table 3.10: Best intra-participant classification performance obtained with different feature sets using TabPFN. Results are reported as mean $\pm$ standard deviation across the seven participants.

3.5 Gait Recognition

Gait recognition was formulated as a closed-set seven-class identification task using walking data from participants P1–P6 and P10. Since all identities must be represented during training, leave-one-participant-out evaluation is not applicable. Instead, the chronological train/validation/test splits generated during windowing were pooled across participants, and a single TabPFN classifier was trained to distinguish between the seven enrolled identities. With seven classes, the chance-level balanced accuracy is approximately 0.143.

Table 3.11 summarizes the overall performance under two preprocessing conditions. Using the raw TENG signals, the classifier achieved a balanced accuracy of 0.986 and a macro-F1 score of 0.985. When normalization Approach 2 was applied, performance decreased to a balanced accuracy of 0.940 and a macro-F1 score of 0.941. Despite this reduction, the normalized model still substantially outperformed the chance baseline.

Participant-wise F1 scores are reported in Table 3.12. For both preprocessing conditions, high performance was observed across all enrolled participants, although larger performance variations were present after normalization.

Model	Normalization	Window [ms] / Overlap [%]	Balanced Acc.	F1 Macro	F1 Weighted
TabPFN	None	3000 / 70	0.986	0.985	0.986
TabPFN	Approach 2	3000 / 90	0.940	0.941	0.929

Table 3.11: Gait recognition performance using the TabPFN classifier and the combined peak-frequency feature set. Results are reported as averages across all participants.

Participant	F1 Macro
P1	0.987
P2	1.000
P3	0.974
P4	0.974
P5	0.988
P6	1.000
P10	0.974

(a) Without normalization

Participant	F1 Macro
P1	0.956
P2	0.928
P3	0.991
P4	0.888
P5	0.860
P6	0.992
P10	0.974

(b) Normalization Approach 2, reference P1

Table 3.12: Participant-wise F1 macro scores for gait recognition.

4 Discussion

4.1 Principal Findings

This work investigated the use of triboelectric nanogenerator (TENG) signals for continuous knee angle estimation, activity recognition, and gait recognition, with a particular focus on inter-subject variability and feature engineering.

For continuous knee angle prediction, excellent intra-participant performance was achieved. Using the proposed Base+Peak+Freq feature set with TabPFN regression, the bike task reached an average R^2 of 0.985 ± 0.015 and a mean absolute error of $2.35^\circ \pm 1.35^\circ$. This demonstrates that TENG signals contain sufficient information to accurately reconstruct knee motion when models are personalized to individual users.

The strongest improvement over previous work of [12] that similarly only used ML models was observed in inter-participant regression. The proposed feature set increased leave-one-participant-out cycling performance to $R^2 = 0.878 \pm 0.118$, substantially exceeding previous classical machine learning results. Since normalization provided little additional benefit for this activity, the improvement appears to be mainly driven by the added peak-morphology and frequency-domain features.

Finally, gait recognition achieved a macro-F1 score of 0.985, showing that TENG walking signals preserve highly distinctive participant-specific characteristics. This supports the conclusion that inter-subject variability is both a challenge for generalization and a potential source of useful biometric information.

Overall, the findings show that TENG-based sensing is highly effective for personalized motion monitoring, and that carefully designed feature representations can substantially improve cross-subject knee angle prediction. At the same time, inter-subject variability, signal quality, and deployment-realistic normalization remain the main obstacles to fully participant-independent motion analysis.

4.2 Comparison with Previous Works

This work builds directly on the two preceding studies in the same laboratory, and results can be compared across all three along the dimensions of regression, classification, and feature engineering. Direct numerical comparison should be treated with some care, however, because participant subsets differ: Spring [12] used six participants with complete data, whereas this work selects seven participants (P1–P6, P8) for regression and a different set of seven (P1–P6, P10) for classification, based on signal quality inspection. Both groups draw from the combined pool recorded by Schaub and Spring.

Intra-participant regression: For the cycling activity, this work achieved $R^2 = 0.985 \pm 0.015$ and $\text{MAE} = 2.35 \pm 1.35^\circ$ using TabPFN with the extended Base+Peak+Freq feature set on unnormalized data. This improves on Spring’s intra-participant result of $R^2 = 0.973 \pm 0.002$ and $\text{MAE} = 3.54 \pm 1.84^\circ$ obtained with the same model but the baseline five-feature set, suggesting that the additional peak-morphology and frequency features carry useful information even in the personalized setting. Schaub [11] reported higher classical ML intra-participant performance (TabPFN/SVR $R^2 \approx 0.97$) and deep learning performance ex-

ceeding $R^2 > 0.98$ with $\text{MAE} \approx 2\text{--}3^\circ$, though those results were obtained with a stricter 2 Hz low-pass filter and a single, well-controlled participant cohort, which likely favoured higher SNR signals.

Inter-participant regression: The most notable improvement over prior work is in the inter-participant setting. With only amplitude based features, classical ML models struggled substantially in Spring’s work (best cycling $R^2 = 0.644 \pm 0.327$ using Random Forest). This work achieves $R^2 = 0.878 \pm 0.118$ and $\text{MAE} = 6.84 \pm 2.97^\circ$ using TabPFN, a substantial gain. Notably, normalized and unnormalized settings produced near identical results for the bike activity, indicating that normalization contributes little additional benefit for cycling specifically. This is consistent with the nature of the activity: the highly repetitive and rhythmic motion of cycling produces more consistent TENG signal characteristics across participants compared to other activities, meaning that amplitude and offset differences between subjects are less of a limiting factor. Another supporting factor for this effect is the consistency of data quality across participants for cycling activity. For example, artifacts such as double peaks were not observed in cycling data. The improvement over Spring is therefore primarily attributable to the extended Base+Peak+Freq feature set, which captures morphological and spectral signal characteristics beyond what the five baseline amplitude features encode. This result approaches the inter-participant deep learning performance reported by Schaub (LSTM $R^2 \approx 0.94$), which is notable because it is achieved with classical ML alone rather than temporal deep learning architectures. The remaining gap may reflect the deeper temporal representations learned by recurrent models, as well as differences in participant cohorts and recording conditions across the two studies.

Intra-participant classification: The extended feature set produces a substantial improvement in intra-participant activity recognition. Spring reported a best balanced accuracy of 0.852 ± 0.092 using TabPFN with the five-feature set and a 1000 ms window. In this work, the same model with Base+Peak+Freq features achieves 0.982 ± 0.038 , an improvement of approximately 13 percentage points. The feature set comparison in Section 3.4 isolates this effect: the base five features alone yield 0.916, adding peak morphology features raises it to 0.964, and the full feature set reaches 0.982. This progression confirms that the peak-morphology and frequency features capture activity-discriminative information that is not encoded in voltage amplitude alone, particularly for distinguishing activities with similar ranges of motion such as squat versus knee bend.

Inter-participant classification: In Spring’s work, inter-participant classification under LOPO barely exceeded the random baseline for a five-class problem, with all models achieving balanced accuracies between 0.480 and 0.504. Without normalization, this work achieves 0.651 ± 0.229 , a meaningful improvement attributable to the richer feature set. Under the activity-specific normalization conditions (Approach 2, P1 reference), balanced accuracy rises to 0.890 ± 0.082 ; however, as noted in Section 3.3.2, this result involves label-dependent normalization that constitutes a form of supervised preprocessing rather than a purely label-agnostic pipeline. The strict, label-agnostic comparison point for this work is therefore the 0.651 result.

Gait recognition: The gait recognition experiment has no direct equivalent in prior work from this laboratory. The near-perfect F1 macro score of 0.985 demonstrates that the TENG walking signal is strongly participant-discriminative, implying that individual gait characteristics are preserved with sufficient fidelity for biometric identification from short windows. This finding is consistent with TENG-based gait identification systems reported in the broader literature [16] and suggests that inter-participant variability, while a challenge for generalization, simultaneously encodes subject-specific information that could be leveraged for user identification or personalized model adaptation.

4.3 Regression

4.3.1 Intra-Participant

The strong intra-participant regression performance, with a mean $R^2 = 0.985$ and $\text{MAE} = 2.35^\circ$ across seven participants, is consistent with the fundamental properties of the cycling activity. Cycling produces

a highly stereotyped, rhythmic knee motion with a consistent range of motion and cadence within each participant’s recording. This means the TENG voltage signal repeats in a regular pattern across hundreds of cycles within a single trial, giving the model ample opportunity to learn a stable voltage-to-angle mapping from the training split. The low within-participant variability of the signal effectively reduces the regression problem to learning a smooth, nearly periodic function, which TabPFN handles well even from a compact feature representation.

The optimal window size of 1000 ms is consistent with this interpretation. At a cycling cadence of approximately 100 rpm, one complete pedaling cycle lasts around 600 ms. A 1000 ms window therefore spans more than one full cycle, providing sufficient temporal context to resolve the instantaneous joint angle from the local signal morphology, while remaining short enough to avoid averaging over phase shifts between cycles.

Per-participant performance is generally high but not uniform. P5 achieves the lowest R^2 of 0.961, while P4 and P8 reach 0.996. Cross-referencing with the SNR analysis in Table 3.1, P5 has a median SNR of 21.84 dB, which is not the lowest in the cohort, suggesting that signal amplitude alone does not fully explain the difference. Residual factors such as subtle differences in sensor placement and individual knee geometry may introduce a degree of non-linearity in the voltage-angle relationship that is harder for the model to capture from handcrafted features alone.

4.3.2 Inter-Participant

The inter-participant regression results reveal both a substantial improvement over prior work and a persistent source of variability across held-out participants. The mean $R^2 = 0.878$ represents a marked gain over Spring’s best classical ML result of 0.644, and this improvement is achieved identically with or without normalization, indicating that the extended feature set captures signal characteristics that transfer better across participants than amplitude-only features.

The most informative aspect of the inter-participant results is the large per-participant spread. P4 and P5 generalize well, with R^2 values of 0.984 and 0.975, while P6 is substantially harder to predict, achieving only $R^2 = 0.698$ and $\text{MAE} = 9.16^\circ$. This heterogeneity can be partly explained by signal quality. P6 has a median SNR of 12.76 dB, the second lowest in the cohort, compared to P4 at 22.30 dB and P5 at 21.84 dB. A low-SNR signal contains more noise relative to the motion-related component, which means that the features extracted from P6’s windows are noisier and therefore harder to predict from a model trained on cleaner signals from other participants. This pattern suggests that improving hardware reliability and sensor placement consistency would have a direct positive impact on inter-participant regression performance.

The fact that normalization yields near-identical results to the unnormalized setting for the cycling activity is an important finding in itself. It indicates that inter-participant amplitude and offset differences are sufficiently small for cycling that they do not constitute the primary bottleneck for generalization. The extended feature set, particularly peak morphology and frequency features, appears to capture the relevant kinematic information in a form that is already sufficiently participant-invariant for this activity.

This contrasts with the classification setting, where normalization does affect performance, and motivates exploring whether normalization becomes more critical for regression on activities with greater signal variability across participants, such as squat or knee bend.

4.4 Classification

4.4.1 Intra-Participant

Intra-participant classification with the extended feature set achieves near-ceiling performance, with a balanced accuracy of 0.982 and F1 macro of 0.983 across five activities. This result reflects both the discrimi-

native power of the TENG signal and the contribution of the extended feature set. The stepwise feature set comparison in Section 3.4 isolates these contributions clearly. The five baseline amplitude features alone achieve a balanced accuracy of 0.916, which is already reasonable but leaves meaningful room for improvement, particularly for activities with overlapping voltage ranges. Adding peak morphology features raises performance to 0.964, and including frequency features brings it to 0.982. Each category therefore contributes non-redundant discriminative information. The amplitude features encode the general signal magnitude associated with each activity’s range of motion. The peak morphology features capture waveform shape, including the timing, prominence, and width of individual contact-separation events, which differ systematically across movement types. The frequency features are particularly valuable for separating cyclic activities such as cycling and walking, which produce periodic signals with a concentrated spectral peak, from discrete, repetitive movements such as squat and knee bend, which are spectrally more diffuse.

The optimal window size of 3000 ms for classification, compared to 1000 ms for regression, reflects the different information requirements of the two activities. Regression requires only the local signal at the window centre to predict the instantaneous angle, while classification requires sufficient signal duration to capture the characteristic temporal pattern of each activity. For slower activities such as squatting, a 3000 ms window is needed to observe at least one complete movement cycle. The window size also affects the performance of peak morphology and frequency features, which are more descriptive in larger windows.

The four-activity evaluation excluding walking yields a slightly lower balanced accuracy of 0.978, which at first appears counterintuitive. The most likely explanation is that the participant groups differ in signal quality between the two evaluations. The walking-excluded group uses P8 instead of P10, and individual differences in model performance across participants, rather than the removal of a confounding class, drive the marginal performance difference.

4.4.2 Inter-Participant

The inter-participant classification results expose both the promise and the current limits of cross-subject generalization. Without normalization, the label-agnostic LOPO evaluation yields a balanced accuracy of 0.651, which represents a meaningful improvement over Spring’s best result of 0.504 and is substantially above the random baseline of 0.20 for a five-class problem. This improvement can be attributed entirely to the richer feature set, since normalization is not applied in this condition, confirming that peak morphology and frequency features encode at least some activity-discriminative information that transfers across individuals.

The high standard deviation of 0.229 indicates that performance is strongly fold-dependent. Some held-out participants are well-predicted by the model while others are near-random. This variability likely reflects how similar a given participant’s signal characteristics are to those of the training set, which is in turn influenced by signal quality, sensor placement consistency, and individual movement execution style.

The choice of reference participant in normalization has a notable effect on results. Using P1 as the reference consistently outperforms P5 across both normalization approaches. This is directly connected to the data quality issues discussed in Section 4.6.2. P5 exhibits double-peak artifacts in squat and knee bend trials, which distort the normalization parameters estimated for those activities. When P5 is used as the reference, these artifacts propagate into the normalized signals of all other participants, degrading the quality of the transformed features for precisely the activities that are already the hardest to classify. Choosing P1 as the reference, whose signal is cleaner and more consistent, avoids this distortion. This finding has the practical implication that in a deployment setting, the reference participant should be selected carefully, based on signal quality.

As discussed in Section 4.6.3, the normalized inter-participant results involve label-dependent preprocessing and cannot be interpreted as strict LOPO performance. However, they are informative in a different sense: they establish an upper bound on what is achievable if activity identity is known at the time of nor-

malization, and they demonstrate that reducing inter-participant amplitude variability through normalization substantially improves separability. A label-agnostic normalization procedure fitted on training participants within each fold and applied without knowledge of the activity label would be the appropriate next step for a deployment-realistic evaluation.

4.5 Gait Recognition

The near-perfect gait recognition performance, with a macro-F1 score of 0.985 and participant-wise F1 scores ranging from 0.974 to 1.000, demonstrates that TENG walking signals contain highly distinctive individual characteristics. This result was achieved using the same feature extraction pipeline and classical machine learning framework employed for the activity classification and regression tasks, without any specialized biometric modelling.

Comparing the normalized and unnormalized experiments in 3.5 provides insight into the information used by the classifier. The macro-F1 score decreased from 0.985 to 0.941 after applying Normalization Approach 2, indicating that participant-specific amplitude characteristics contribute to identity recognition. However, the normalized model still performed substantially above the chance level of 0.143, suggesting that gait-dependent signal patterns alone remain highly informative for distinguishing individuals.

These findings can be viewed as the inverse of the inter-participant generalization problem encountered throughout this work. The same differences in anatomy, sensor placement, and movement execution that hinder generalization across participants also make individuals distinguishable from one another. Consequently, TENG signals encode both activity-related and identity-related information. This observation suggests that participant-adaptive approaches, in which a small amount of user-specific calibration data is incorporated into a general model, may offer a promising direction for improving inter-participant classification and regression performance.

A limitation of the current evaluation is the train/validation/test splitting procedure. The chronological split is performed at the window level after trial concatenation, meaning that a small number of adjacent windows may lie on opposite sides of a split boundary. Because overlapping windows share a portion of their underlying samples, this introduces a limited amount of information leakage between subsets. However, the effect is restricted to the immediate vicinity of the split boundaries and affects only a very small fraction of the evaluation windows. Given the substantial margin between the reported performance and chance level, it is unlikely that this leakage is a major contributor to the observed gait recognition results.

4.6 Limitations

4.6.1 Small Cohort Size and Data Imbalance

The data used in this study was collected from 11 participants, however, some of them have missing or incomplete data. Furthermore, some portion of the existing data is unusable due to poor quality. For this reason, most of the results were reported only for 7 participants, which is a very limited dataset. This small size limits the generalization capability of the model to a wider range of users.

4.6.2 Data Quality Limitations

One of the greatest limitations encountered in this study was inconsistencies in data quality the experiments demonstrated how strongly signal quality influenced both regression and classification performance. Importantly, the observed quality issues were not limited to high-frequency noise that could be removed through filtering. Instead, many artifacts altered the underlying structure of the TENG signal itself and therefore affected the extracted features used for machine learning.

As it was pointed out by [12], some TENG signals had two sets of peaks corresponding to each knee motion or ground truth angle. Ideally, one contact-separation event between the triboelectric layers should occur per movement cycle, resulting in one positive and one negative voltage peak. The presence of additional peaks suggests that the sensor layers made contact multiple times during a single motion. The causes for this phenomenon could be differences in individual anatomy, variation in sensor placement or loosening of the knee sleeve containing the TENG sensor over the course of measurements. The occurrence of the "double peak" artifact was not consistent across participants, activities, or even different trials of the same activity for one participant. This last observation strengthens the hypothesis that the double peaks occurred due to the loosening of the knee sleeve. Double peaks were observed most frequently during squat and knee bend activities, which involve larger ranges of motion and more pronounced deformation of the knee sleeve.

The double-peak artifact directly affects the extracted signal characteristics. Peak-based features may incorrectly interpret the additional peaks as separate movement events, while frequency-domain features may indicate artificially increased movement frequencies. Consequently, the same physical movement can produce substantially different feature representations across recordings, reducing model robustness and limiting inter-participant generalization.

Another issue observed was the polarity inconsistency of the TENG data. Normally, a TENG signal cycle should contain a positive peak followed by a negative peak. However, in some of the data it was observed that this polarity was flipped, one cycle consisted of a negative peak followed by a positive peak. Such cases likely occurred because the sensor's connections were reversed during data collection. Although this is an easily corrected issue during the manual alignment correction stage, identifying polarity inversions becomes increasingly difficult when the signal quality is already poor or when the movement pattern is irregular. Such inconsistencies introduce additional variability unrelated to the actual motion being measured.

4.6.3 Normalization Limitations

The proposed normalization approaches were developed to reduce inter-subject variability and improve generalization across participants. While the results demonstrated substantial improvements in inter-participant classification performance, several limitations remain.

One limitation of the proposed normalization approaches is that the normalization parameters were estimated using the complete recording of each participant. In a real-world deployment scenario, future samples would not be available at the time normalization is performed. Consequently, the current implementation introduces a degree of information leakage by using statistics derived from the entire signal. To investigate the practical impact of this issue, calibration-window experiments were conducted for both normalization approaches. The results showed that parameters estimated from only a short initial segment produced normalized waveforms that were highly similar to those obtained using the full recording. Therefore, while the current implementation uses the complete recording to estimate normalization parameters, the results indicate that comparable normalization can likely be achieved using only a short calibration segment. The use of full-recording information should therefore be interpreted as a methodological simplification rather than a fundamental requirement of either normalization approach.

A more important limitation of both normalization approaches is the introduction of **class-label leakage** in the activity classification task. In the current implementation, normalization parameters are estimated separately for each activity and each window is normalized using the transform corresponding to its true activity label. Consequently, the preprocessing stage already uses information that would not be available for an unseen sample in a real deployment setting. The normalized inter-participant classification results should therefore be interpreted as optimistic estimates rather than strict label-agnostic LOPO performance.

This leakage likely contributes to the substantial performance increase observed after normalization, with macro-F1 scores reaching 0.875 ± 0.095 . By reducing activity-specific amplitude differences relative to a reference participant, the normalization procedure increases class separability in a manner that would

not be possible without prior knowledge of the true activity.

An additional observation is that Approach 2 consistently outperformed Approach 1 despite both approaches relying on the same activity-specific normalization framework and therefore being affected by the same class-label leakage. This suggests that the performance difference cannot be explained solely by leakage and is also related to how the two normalization procedures modify the underlying TENG waveform. A possible explanation for the performance difference is as follows.

The voltage signal of a participant can be expressed as a participant-specific linear relationship with knee angle plus a residual component,

$$V_p(t) = a_p\theta(t) + \epsilon_p(t) + b_p, \quad (4.1)$$

where $\epsilon_p(t)$ describes the signal characteristics that are not explained by the linear voltage–angle relationship. Applying Approach 1 yields

$$V_{\text{scaled}}(t) \approx a_{\text{ref}}\theta(t) + \frac{a_{\text{ref}}}{a_p}\epsilon_p(t) + b_{\text{ref}}. \quad (4.2)$$

Consequently, the residual component is amplified or attenuated by the factor a_{ref}/a_p , which varies substantially across participants and activities, due to noise or recording artifacts. In some participant–activity combinations, the fitted slope a_p is close to zero, resulting in particularly large scaling factors.

In contrast, Approach 2 applies a scale factor derived from stable signal statistics and therefore it is less susceptible to noise amplification. This preserves the relative shape of the waveform even for the noisy cases and therefore maintains the consistency of peak morphology and frequency-domain features across participants.

This interpretation is consistent with the confusion matrices in Figure 3.6, where the largest improvements of Approach 2 appear in squat, knee bend, and walking activities, which rely strongly on repetitive signal structure and peak morphology. In comparison, biking, which is characterized by a more regular and amplitude-dominated signal pattern, exhibits comparatively smaller improvements. These observations suggest that preserving the intrinsic waveform structure may be more beneficial for activity recognition than enforcing agreement with a participant-specific voltage–angle relationship.

Overall, the normalized classification results demonstrate the potential benefit of reducing inter-participant variability, but they should not be interpreted as deployment-realistic performance. A more rigorous evaluation would require an activity-agnostic normalization procedure that can be applied without access to the true activity label.

4.7 Future Work

Several directions can be explored to further improve the applicability of TENG-based motion monitoring systems.

One of the most important areas for future work is the development of activi-agnostic normalization methods. While the normalization approaches proposed in this work successfully reduced inter-participant signal variability, their current implementation depends on activity-specific transformations. Future approaches should estimate normalization parameters using only training participants and apply them to unseen users without requiring knowledge of the activity being performed.

Further improvements may also be achieved through hardware refinements and more consistent sensor placement. A significant portion of the observed inter-participant variability appears to originate from differences in anatomy, sleeve positioning, and sensor contact conditions. Improved fixation mechanisms or alternative sensor designs may reduce artifacts such as double peaks and improve signal consistency across users.

Although the proposed peak morphology and frequency-domain features substantially improved inter-participant regression performance, more advanced machine learning approaches could be investigated. In particular, subject-invariant representation learning, domain adaptation techniques, and deep learning models operating directly on raw TENG signals may further improve generalization to unseen individuals.

Finally, future studies should include larger and more diverse participant cohorts. A larger dataset would enable more reliable evaluation of participant-independent models and provide a better understanding of how factors such as age, body composition, anatomy, and movement style influence TENG signal characteristics. Such studies would be essential for assessing the feasibility of deploying TENG-based motion monitoring systems in real-world applications.

5 Conclusion

This work investigated the use of triboelectric nanogenerator (TENG) signals for continuous knee angle estimation, human activity recognition, and gait recognition. Building upon previous work in the Biomedical and Mobile Health Technology Lab, two normalization approaches and several new feature representations were introduced and evaluated using the TabPFN foundation model.

The results demonstrate that TENG signals contain rich biomechanical information suitable for both regression and classification tasks. In the personalized setting, knee angle estimation achieved excellent performance, reaching ($R^2 = 0.985$) with a mean absolute error of only (2.35°). Activity recognition likewise achieved near-perfect performance within individual participants, confirming that the proposed feature representations successfully capture task-specific characteristics of the TENG signal.

The most important contribution of this work is the substantial improvement in inter-participant knee angle regression. Compared with the previous ML-based approach, the achieved LOPO performance demonstrates that richer feature representations can significantly improve cross-subject regression. By augmenting the original feature set with peak morphology and frequency-domain descriptors, leave-one-participant-out performance increased to ($R^2 = 0.878$). This finding suggests that the limitations observed in earlier studies were not solely due to the sensing modality itself but also to the information content of the extracted features.

The inter-participant activity classification task also benefited from the richer feature set. An improvement over the previous works were observed in true LOPO performance. With the help of proposed normalization approaches, great performance gain was achieved; however, this gain was likely influenced by the label leakage arising from task-specific normalization. Future work should therefore focus on task-agnostic normalization procedures that can be applied to unseen participants without requiring activity labels.

The gait recognition experiments revealed an additional important characteristic of TENG signals: they preserve highly distinctive subject-specific information. While this property enables near-perfect biometric identification, it also highlights the fundamental challenge of inter-participant generalization. The same individual differences that make people identifiable from their gait also make it more difficult to build participant-independent activity recognition and regression models.

In conclusion, the findings of this study demonstrate that TENG-based sensing is a promising technology for self-powered wearable motion monitoring. The strong personalized performance and improved cross-subject regression results indicate considerable potential for applications in rehabilitation, biomechanics, and long-term health monitoring. At the same time, further advances in sensor design, data quality, normalization strategies, and participant-independent modelling will be necessary before robust deployment across large and diverse user populations becomes feasible.

A The First Appendix

A.1 Metrics and Definitions

A.1.1 Regression Metrics

The coefficient of determination (R^2) measures the proportion of variance in the ground-truth values explained by the model predictions:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (\text{A.1})$$

where y_i is the ground-truth value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the ground-truth values, and N is the number of samples.

The mean absolute error (MAE) is defined as

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (\text{A.2})$$

The root mean squared error (RMSE) is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}. \quad (\text{A.3})$$

A.1.2 Classification Metrics

Classification accuracy is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (\text{A.4})$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively.

For multi-class problems with class imbalance, balanced accuracy was used. It is defined as the average recall across all C classes:

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c}. \quad (\text{A.5})$$

Precision and recall for class c are defined as

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad (\text{A.6})$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}. \quad (\text{A.7})$$

The F1-score for class c is the harmonic mean of precision and recall:

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (\text{A.8})$$

Macro-averaged F1-score assigns equal weight to each class and is computed as

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c. \quad (\text{A.9})$$

Weighted F1-score accounts for class frequencies:

$$F1_{\text{weighted}} = \sum_{c=1}^C \frac{n_c}{N} F1_c, \quad (\text{A.10})$$

where n_c is the number of samples belonging to class c and N is the total number of samples.

A.2 Data

The following table shows the data status for each participant and activity type as reported by [12].

Participant	Bike	Squat	Knee Bend	Side Lift	Walk
P1	c	c	c	c	q
P2	c	c	c	c	c
P3	c	c	c	c	i
P4	c	c	c	c	c
P5	c	c	c	c	c
P6	c	c	c	c	i
P7	m	c	m	c	m
P8	c	c	c	c	m
P9	c	c	c	c	m
P10	c	c	c	c	i
P11	c	n	m	c	m

Table A.1: Overview of data status per participant and task. c = complete, i = incomplete (one repetition could not be reconstructed due to missing markers, the data could still be used for the analysis), n = not recorded (task was not recorded), a = there was no signal generated by this motion, q = insufficient quality (recording is only noise, not meaningful), m = markers missing (recording could not be reconstructed due to missing markers, which the cameras did not capture as the participant was too close to the wall).

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